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NeuroEvolve: A brain-inspired mutation optimization algorithm for enhancing intelligence in medical data analysis

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ABSTRACT

Medical data analysis often suffers from high dimensionality, noise, and complex non-linear patterns, causing traditional optimization-based learning methods to converge slowly and leads to suboptimal predictions. To address these challenges, we propose NeuroEvolve, a brain-inspired mutation strategy integrated into Differential Evolution (DE) that dynamically adjusts mutation factors based on feedback, enhancing both exploration and exploitation. The method was evaluated on three benchmark medical datasets – MIMIC-III, Diabetes, and Lung Cancer – using standard performance metrics including Accuracy, F1-score, Precision, Recall, and a newly defined Mean Error Correlation Coefficient (MECC). Comparative experiments with state-of-the-art evolutionary optimizers demonstrate NeuroEvolve's superiority. For instance, on the MIMIC-III dataset, NeuroEvolve achieved an Accuracy of 94.1% and an F1-score of 91.3%, reflecting an improvement of 4.5% in Accuracy and 6.2% in F1-score over the best-performing baseline Hybrid Whale Optimization Algorithm (HyWOA). Similar improvements were consistently observed across the Diabetes and Lung Cancer datasets, confirming the robustness of the our approach. These results highlight the potential of NeuroEvolve as an effective optimization-driven learning strategy for complex medical prediction tasks.

1. Introduction

The exponential growth of medical data, propelled by electronic health records (EHRs), advanced imaging technologies, and pervasive sensor-based monitoring, presents an immense opportunity to enhance clinical diagnosis, refine treatment planning, and ultimately improve patient outcomes (Ismailova et al., 2026; Li et al., 2016). However, fully realizing this potential remains a significant challenge. Medical datasets are inherently heterogeneous, often high-dimensional, and frequently noisy, which makes their processing with traditional algorithms particularly difficult (Anari et al., 2024). Furthermore, existing optimization methods, such as Genetic Algorithms (GA), Differential Evolution (DE), and Particle Swarm Optimization (PSO), commonly suffer from issues like premature convergence, static parameterization, and limited adaptability. These drawbacks often lead to suboptimal predictions, especially in dynamic healthcare environments (Cheng, Zhou, Hu, Mohamed, & Liu, 2023; Gao et al., 2019). Such limitations underscore the critical need for more adaptive optimization strategies capable of capturing the evolving and nonlinear nature of complex medical data.

Various optimization approaches have been explored in medical data analysis, including the aforementioned GA, DE, and PSO (Cheng et al., 2023; Gao et al., 2019). More recently, hybrid methods, such as hybrid DE (HyDE), Grey Wolf Optimizer (GWO)-based variants, and Whale Optimization Algorithm (WOA)-based models, have sought to improve convergence and robustness by combining complementary search strategies (Anari, Sadeghi, Sheikhi, Ranjbarzadeh, & Bendechache, 2025; Kia, 2025). Optimization has also been effectively applied to specific healthcare tasks, such as image fusion for enhancing diagnostic clarity (Kavita, Alli, & Rao, 2022) and feature selection for breast cancer diagnosis utilizing metaheuristics with soft voting classifiers (Benghazouani, Nouh, & Zakrani, 2025; Pineda, Valencia-Arias, Giraldo, & Zapata-Ochoa, 2026). While these existing approaches

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have demonstrated promising improvements, they often rely on fixed or heuristically updated parameters. This reliance significantly limits their effectiveness in handling dynamic and heterogeneous medical datasets. Crucially, most current methods prioritize accuracy while often neglecting clinically vital measures, including recall and correlation with actual outcomes. This oversight creates a clear research gap for adaptive frameworks that can effectively balance exploration, exploitation, and clinical relevance. Our proposed NeuroEvolve framework is specifically designed to address this critical gap.

1.1. Highlights of problem description

Despite the increasing availability of complex clinical datasets, their full potential remains significantly underutilized due to several persistent challenges:

1. **Heterogeneity:** Medical data originate from diverse sources, including structured, semi-structured, and unstructured formats (Gao et al., 2019; Zenbout, Bouramoul, Meshoul, & Amrane, 2023), making unified processing difficult.
2. **Dynamism:** Evolving medical practices and patient conditions mean static algorithms often fail to capture critical, temporal changes (Sherry et al., 2001).
3. **Generalizability Risks:** There is a high risk of overfitting and poor generalizability across varied patient populations and disease types (Alharbi & Vakanski, 2023; Mirjalili, Saremi, Mirjalili, & Coelho, 2016).
4. **Clinical Irrelevance:** Models frequently produce mathematically accurate but clinically irrelevant or uninterpretable outputs (Anari et al., 2025).

To effectively overcome these significant challenges, we propose NeuroEvolve, an adaptive optimization framework that integrates a brain-inspired mutation strategy within the Differential Evolution (DE) paradigm. Distinct from fixed-rate algorithms, NeuroEvolve leverages principles akin to neuroplasticity, enabling it to dynamically adjust its mutation behavior based on real-time performance feedback. This inherent adaptability significantly improves feature relevance, enhances convergence stability, and strengthens clinical decision support across various tasks, including diagnosis, prognosis, and treatment optimization. While existing hybrid bio-inspired methods (e.g., HyDE, GWO variants) have made strides in optimization, very few have explored feedback-driven cognitive adaptation specifically for medical data analysis (Anari et al., 2025; Kia, 2025). NeuroEvolve bridges this critical gap, offering an optimization-driven learning strategy that markedly enhances prediction accuracy, stability, and generalizability, thereby supporting earlier diagnosis and improved patient care.

1.2. Contributions of the paper

This study makes the following key contributions:

- **Novel Brain-Inspired Mutation Strategy:** We introduce a dynamic mutation operator within the DE framework that effectively mimics neuroplasticity. This innovation enables real-time adaptability and significantly improved convergence in handling high-dimensional medical datasets.
- **Proposed Evaluation Metric (MECC):** A new metric, the Mean Error Correlation Coefficient (MECC), is developed to jointly evaluate prediction accuracy and error correlation. This provides a more comprehensive and robust performance assessment than conventional metrics alone.
- **Comprehensive Experimental Validation:** The proposed *NeuroEvolve* algorithm is rigorously evaluated on three benchmark medical datasets (MIMIC-III, Diabetes, and Lung Cancer) using a suite of standard metrics including Accuracy, F1-score, Precision, Recall, and the newly defined MECC.
- **Significant Performance Improvements:** Results conclusively demonstrate that *NeuroEvolve* consistently outperforms established baseline optimizers (HyDE, GA, HyGWO, HyWOA). For instance, it achieved up to 4.5% higher Accuracy and 6.2% higher F1-score on the MIMIC-III dataset, with similar notable gains consistently observed across the other datasets.
- **Enhanced Clinical Relevance and Robustness:** By integrating dynamic adaptability and interpretability into its design, our approach substantially enhances prediction robustness and generalizability. This highlights its significant potential for practical application in real-world healthcare decision support systems.

The remainder of this paper is organized as follows: Section 2 provides a review of related work; Section 3 presents the mathematical formulation of our approach; Section 4 describes the proposed methodology in detail; Section 5 reports and analyzes the experimental results; and Section 6 concludes the paper, offering insights for future research.

2. Literature review

The field of medical dataset analysis has witnessed significant research focused on enhancing various tasks related to diagnosis, prognosis, and decision-making (Kia et al., 2025). This section introduces the reader to several evolutionary algorithms relevant to the general framework of this article, highlighting how these techniques contribute to improving medical dataset analysis.

Research in Alharbi and Vakanski (2023) underscores the necessity of incorporating local search techniques into DE algorithms to improve their applicability across different optimization problems. Similarly, Saravanan and Madheswaran (2014) addresses premature convergence in GAs, proposing modifications that enhance their performance and prevent early convergence in continuous optimization problems. In Mirjalili et al. (2016), authors employed DE alongside homeostasis-based mutation approaches for solving multi-objective optimization problems, while other methods for improving DE's performance in multi-objective scenarios have also been presented (Tawhid & Ibrahim, 2020). In the medical domain, considerable progress has led to practical applications, such as genome-based cancer prediction and diagnosis, where machine learning methods have demonstrated high precision (Genome data set, 2020).

The GWO, a bio-inspired metaheuristic algorithm based on the social organization of grey wolf populations, was utilized in Mokoatle, Marivate, Mapiye, Bornman, Hayes, et al. (2023) for optimization problems. Algorithms that mimic the cooperative hunting behavior of wolves have yielded promising results in optimization and search problems, particularly when applied to health information systems and medical data (Singh, Singh, & Mehta, 2021). Furthermore, deep learning strategies, notably CNNs, exhibit strong potential in differentiating between malignant and benign nodules in radiological examinations (Guo et al., 2023). This highlights the extensive possibilities for data fusion and advanced artificial intelligence in cancer diagnosis.

In recent research by Guo et al. (2023) (Singh, 2017), machine learning algorithms were successfully employed to predict prostate cancer progression using clinical and genetic features, achieving noteworthy sensitivity and specificity (Genome data set, 2020). This work emphasizes the application of personalized medicine for individual treatment strategies and cancer predictions. Complementary studies have utilized artificial neural networks (ANNs) to predict ovarian cancer survival rates (Chen, Li, Li, Zhao, & Dong, 2023), and gene expression data from microarray studies to increase the efficiency of identifying malignant tumors (Kumar & Garg, 2023). Collectively, these studies demonstrate the expanding role of machine learning in medicine, particularly in cancer prediction, detection, and prognosis. Moreover, the application of various deep learning methods, including CNNs, recurrent neural networks (RNNs), and variational autoencoders, has proven effective in extracting intricate patterns from genomic data, thereby

Table 1
Description of notation and abbreviation.

Notation	Description
G	Number of Generation
Cr	Crossover Rate
δ_1	Scale factor of mutation operator
D	Dimensions
$MaxFunEvs$	Number of Function Evaluation
P	Population
θ	Brain-Inspired Parameters
S_x	Mutation Strategy for Individual x
x_1, x_2, x_3, x_4	Distinct Individuals for Mutation
u_x	Mutant Vector for Individual x

improving cancer diagnosis and prognosis (Lee, Park, & Lee, 2022; Leng et al., 2022).

Additional advancements in optimization include the use of an advanced learning-based Brain Storm Optimization (BSO) algorithm in Fu, Wang, Gao, Suganthan, and Huang (2024) to solve complex scheduling and routing problems in manufacturing systems. This exemplifies how metaheuristics can effectively manage multi-objective and constraint-heavy industrial optimization challenges. Similarly, Fu, Ma, Gao, Li, and Dong (2024) describes a modified artificial bee colony algorithm applied to the real-world complexities of routing and scheduling in home healthcare services, illustrating metaheuristics' applicability in logistics under uncertainty and with multiple conflicting objectives. Integrating multi-omics data with graph active learning-based approaches further improves the generalization of these models, enabling more precise individual predictions (Sarshar et al., 2024). Overall, these diverse techniques offer a promising pathway for promoting precision medicine, enhancing the effectiveness of healthcare solutions, and advancing cancer treatments. This article specifically aims to offer insights for improved cancer prediction, focusing on whole-genome deep learning and personalized cancer genomics.

The aforementioned body of work reveals several research gaps that inspire the current study on intelligent mutation-based evolutionary optimization algorithms, particularly within genomics and personalized medicine (Chen et al., 2023; Kumar & Garg, 2023; Ramirez et al., 2021; Tawhid & Ibrahim, 2020). The identified gaps warranting further investigation include:

1. **Scalability Challenges:** A significant limitation pertains to evaluating algorithm performance in processing extensive genomic data, especially high-dimensional genetic profiles. The increasing size of genomic datasets presents a critical scalability issue for optimization algorithms tuned for data processing.
2. **Disease Generalizability:** Further research is needed to assess the algorithm's performance across various diseases and their subtypes. Such studies are crucial for determining the proposed algorithm's efficacy in addressing diverse precision medicine problems.
3. **Comparative Validation:** Rigorous validation of the new algorithm's effectiveness is required through comprehensive comparisons with other advanced optimization algorithms and established machine learning tools. This will provide a clearer understanding of the proposed approach's specific strengths in genomics and innovative health applications.
4. **Domain-Specific Integration:** Incorporating explicit biological and medical domain knowledge into the optimization process can significantly improve algorithm results. Further research should explore the feasibility and impact of utilizing domain-specific constraints.

3. Mathematical formulation

To address the challenges posed by genomics and precision medicine applications (Chen et al., 2023; Gao et al., 2019; Kumar & Garg, 2023; Ramirez et al., 2021; Shi, Zhang, & Wang, 2023), we introduce an optimization-based approach. The model's accuracy is significantly improved by incorporating extensive genetic data and other patient-relevant traits. For example, the learning process is guided by factors such as ten genetically determined mutations, clinical parameters, and a patient's medical history. This integration enhances the specificity of model predictions, facilitating personalized therapy planning and accurate evaluation of disease outcomes. To maximize prediction accuracy on the training data, the training algorithm optimizes the loss function by minimizing its value with respect to the model parameters. Feature significance analysis and attention mechanisms are employed to interpret these predictions, ensuring the model remains transparent and comprehensible. Table 1 lists the notations and abbreviations used in this work. The proposed model's significance lies in its ability to leverage the DE algorithm for managing complex genomic data while integrating patient-specific traits, which is crucial for personalized medicine applications. The model is formally described as follows:

1. Input Data: Let X denote the input genomic data matrix, where each row corresponds to a patient (sample) and each column represents a genetic feature (gene). The element X_{ij} represents the expression level or genetic variation of gene j in sample i.

2. Target Variable: Let Y represent the vector of target variables for each patient, such as disease outcomes or treatment responses. The element Y_i indicates the target variable for sample i.

3. Model Parameters: Let θ be a vector of model parameters that are learned from the input data X to predict the target variable Y. The parameter vector is defined as $\theta = [\theta_1, \theta_2, \ldots, \theta_n]$, where n is the number of parameters.

4. Predictive Model: The predictive model maps the input genomic data X to the target variable Y using the learned parameters θ. This relationship is expressed as:

$$Y = f(X, \theta), \tag{1}$$

where $f(X, \theta)$ represents the model function.

5. Model Training: The training process involves optimizing the parameters θ to minimize the prediction error between the predicted values $\hat{Y}$ and the actual target values Y. This optimization problem is expressed as:

$$\theta^* = \arg\min_{\theta} L(Y, f(X, \theta)), \tag{2}$$

where $L(\cdot)$ denotes the loss function. Commonly used loss functions include Mean Squared Error (MSE) for regression and cross-entropy loss for classification.

6. Precision Medicine Application: For precision medicine, the model parameters θ may incorporate patient-specific features, such as genetic mutations or clinical variables, enabling personalized treatment. This is formalized as:

$$Y_i = f(X_i, \theta_i), \tag{3}$$

where θ_i captures the individual characteristics of patient i influencing the treatment response or disease outcome.

7. Prediction Model: The proposed linear regression model for predictions is expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon, \tag{4}$$

where y is the predicted output, $x_1, x_2, \ldots, x_n$ are the input features, $\beta_0, \beta_1, \beta_2, \ldots, \beta_n$ are the model coefficients to be learned, and ε is the error term representing the discrepancy between predicted and actual values.

The objective of training is to find the coefficients $\beta_0, \beta_1, \beta_2, \ldots, \beta_n$ that minimize the error between predicted and actual values in the training data. Optimization techniques such as least squares regression are typically employed for this purpose. Once trained, the model can be used to predict new outputs for unseen input data by substituting the feature values $x_1, x_2, \dots, x_n$ into the model.

8. Mean Error Correlation Coefficient (MECC)

The MECC is a custom metric to assess the relationship between prediction errors and correlation in a dataset. It combines two distinct measurements: Mean Error (ME) and Correlation Coefficient (CC). Mathematically, MECC is defined as in Eq. (5), where ME represents the Mean Error, which calculates the average prediction error in a dataset. The term CC represents the Correlation Coefficient, quantifying the linear relationship between two variables. The MECC metric effectively combines these two components, assigning more weight to high correlation and low prediction errors. A higher MECC value indicates a stronger correlation and lower prediction errors. By integrating Mean Error and the Correlation Coefficient into MECC, we establish a customized metric that emphasizes accuracy and correlation in our analysis or modeling process. Mathematically, MECC is defined as:

$$\text{MECC} = \frac{ME \times CC}{|ME| + |CC| + \epsilon} \tag{5}$$

Where:

- ME is the Mean Error, calculated as:

$$ME = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{y}_i \right) \tag{6}$$

  with y_i being the true value and $\hat{y}_i$ the predicted value.
- CC is the Correlation Coefficient between true and predicted values:

$$CC = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2 \sum_{i=1}^{n} (\hat{y}_i - \bar{\hat{y}})^2}} \tag{7}$$

- ϵ is a small constant to prevent division by zero.

MECC combines both error magnitude and correlation strength, offering a balanced measure of model accuracy and reliability. It ensures models used for medical data not only minimize errors but also maintain a strong correlation with ground truth, crucial for reliable predictions.

9. Fitness Function The fitness function for smart healthcare applications, calculated in Eq. (8), combines all objectives from Eqs. (1)–(5) into a single objective function. Here, fun_1, fun_2, fun_3, fun_4, and fun_5 represent the weights assigned to each objective function.

$$\begin{aligned} Fitness = {} & fun_1 \times Eq.\ (1) + fun_2 \times Eq.\ (2) + fun_3 \times Eq.\ (3) \\ & + fun_4 \times Eq.\ (4) + fun_5 \times Eq.\ (5) \end{aligned} \tag{8}$$

4. Proposed methodology

To extend the DE algorithm with brain-inspired principles, we propose the *NeuroEvolve-DE* approach for enhancing medical data processing. This innovative extension enriches DE with neuron-like mutation mechanisms, enabling improved decision-making in healthcare and fostering further advancements in medical applications. The core of this approach is the *Brain-Inspired Mutation-based DE* (BI-DE) technique, which builds upon the traditional DE framework by introducing mutation operators that mimic cognitive processes and neural adaptation. These operators allow the algorithm to dynamically adjust its mutation strategies in real-time, akin to how the human brain devises novel solutions to complex problems.

By integrating domain-specific constraints and knowledge, our approach provides fine-grained control to optimize performance across diverse and multifaceted solution spaces. Furthermore, the algorithm incorporates mechanisms for tuning mutation parameters, thereby balancing the exploration and exploitation phases of the search process to achieve optimal convergence. This method retains the explorative power of DE while guiding the optimization using heuristics inspired by neuronal design. The subsequent sections present a detailed, step-by-step description of the proposed algorithm.

4.1. The NeuroEvolve-DE: Initialization of the brain-inspired optimization algorithm

Like other DE algorithms, the proposed NeuroEvolve-DE algorithm begins with an initialization phase. This phase lays the foundation for subsequent iterations by generating an initial population P, consisting of candidate solutions, and defining key parameters that influence the algorithm's performance. The population is designed to represent various scenarios encountered in medical dataset analysis, allowing the algorithm to be tailored to specific contexts and enhancing its convergence speed.

Inspired by the human brain's adaptability and cognitive learning mechanisms, NeuroEvolve-DE introduces brain-inspired parameters, denoted as θ, which guide the optimization process. These parameters dynamically influence the mutation and selection strategies at each iteration, simulating learning-driven adjustments analogous to synaptic plasticity.

At each iteration, the algorithm evaluates the fitness of each individual in the population to assess its quality relative to the optimization objective, as detailed in Algorithm 1. The best-performing individuals are preserved, allowing their advantageous traits to propagate through the population. This evolutionary process continues until predefined convergence criteria are satisfied, such as reaching a maximum number of iterations or achieving a target solution quality. Once these conditions are met, the algorithm selects and returns the best solution from the final population, denoted by x^*, as the optimal result. This brain-inspired evolutionary framework enables NeuroEvolve-DE to dynamically adapt to changing data patterns and problem complexities, offering enhanced performance and robustness in medical data analysis tasks.

Algorithm 1: Brain-Inspired Optimization Algorithm

```
Data: Population P, Brain-Inspired Parameters θ, Termination
      Criteria
Result: Optimal Solution x*

Initialize a random population P with brain-inspired
  parameters θ;
while Termination criteria are not met do
    foreach x in P do
        Apply brain-inspired operations to x based on θ;
    end
    Evaluate the fitness of each solution in P;
    Select the best-performing solutions based on fitness;
    Update the population P with the selected solutions;
end
Identify and return the optimal solution x* from the final
  population P;
```

4.2. Brain-inspired mutation-based approach

The brain-inspired mutation operator incorporates the concept of neuroplasticity—the brain's ability to adapt by reorganizing its neural connections in response to stimuli. Translating this concept into our optimization framework, the mutation operator dynamically adjusts its step size and direction based on both the fitness landscape and the historical performance of the population. The mutation process is mathematically formulated as follows:

$$X_i^{t+1} = X_i^t + \alpha \cdot N(0, \sigma^2) + \beta \cdot (X_b^t - X_i^t) \tag{9}$$

Where: X_i^t is the solution vector of the ith individual at generation t. $N(0, \sigma^2)$ represents a Gaussian distribution with mean 0 and variance σ^2, simulating random noise akin to synaptic weight adjustments in neural networks. The parameter α is a dynamic scaling factor that adapts based on the individual's fitness improvement rate, encouraging exploration if progress stagnates. The parameter β is an influence coefficient guiding the solution towards the global best X_b^t, mirroring how neural networks reinforce connections based on feedback. Finally, X_i^{t+1} is the updated solution after mutation. The fitness-based scaling factor (α) is mathematically formulated as:

$$\alpha = \exp\left(-\gamma \times \frac{f(X_i^t) - f(X_b^t)}{f(X_b^t)}\right) \tag{10}$$

where γ is a sensitivity parameter and $f(\cdot)$ denotes the fitness function. This mechanism uses Gaussian perturbation to mimic synaptic fluctuations, ensuring a balance between exploration and exploitation. The neuro-inspired adjustment, controlled by β, directs the search based on the global best solution, simulating how neurons strengthen or weaken synaptic connections in response to learned experiences.

The proposed Brain-Inspired Mutation-based algorithm, outlined in Algorithm 2, is designed to be highly suitable for complex optimization problems. It begins with the generation of a random population P and the integration of brain-inspired parameters θ. These parameters are critical guidelines that assist in the adaptation of the mutation process. The primary objective is to ensure that by the final generation, the population P has converged to the optimal solution x^*.

Algorithm 2: Brain-Inspired Mutation-based Algorithm

Data: Population P, Brain-Inspired Parameters θ, Termination Criteria
Result: Optimal Solution x^*

Initialize a random population P with brain-inspired parameters θ;
while *Termination criteria are not met* **do**
  foreach x *in* P **do**
    Select a mutation strategy S_x based on θ;
    Randomly select four distinct individuals x_1, x_2, x_3, x_4 from P for mutation;
    Generate the mutant vector u_x using the selected strategy and x_1, x_2, x_3, x_4;
    Apply domain-specific constraints to u_x;
    Evaluate the fitness of u_x and x;
    if *Fitness of u_x is better than x* **then**
      Replace x with u_x in P;
    end
  end
end
Identify and return the optimal solution x^* from the final population P;

In the proposed algorithm, each solution x in the population P undergoes a mutation process guided by a strategy S_x, which is selected based on the brain-inspired parameters θ. To create a mutant vector u_x, the algorithm randomly selects four distinct individuals (x_1, x_2, x_3, x_4) from the population P. The chosen mutation strategy S_x is then applied to these individuals to generate u_x. The algorithm ensures that u_x adheres to any domain-specific constraints, maintaining its validity within the problem space. Subsequently, the fitness of the mutant vector u_x is compared to that of the original solution x. If u_x exhibits superior fitness, it replaces x in the population. This selection step ensures the retention of high-performing solutions, promoting the algorithm's convergence toward an optimal result. This iterative process continues until the specified termination criteria are met.

4.3. Proposed brain-inspired mutation-based DE (BI-DE) algorithm

Algorithm 3 presents an optimization strategy that integrates brain-inspired mutation mechanisms into an enhanced DE framework. It begins by initializing a random population P and the mutation parameters θ. In each iteration, the algorithm employs a mutation strategy for every solution in P, selected from a predefined set of strategies based on θ. A mutant vector u_x is generated by randomly selecting three distinct individuals from the population to serve as donor vectors. The algorithm then evaluates the fitness of both the original solution and the mutant vector. If the mutant vector has better fitness, it replaces the original solution in the population. Additionally, crossover and selection operators can be applied to enhance the diversity of the population. This process repeats until the termination conditions are satisfied, at which point the optimal solution x^* is returned from the final population.

Algorithm 3: BI-DE Algorithm

Data: Population P, Brain-Inspired Parameters θ, Termination Criteria
Result: Optimal Solution x^*

Initialize a random population P with brain-inspired parameters θ;
while *Termination criteria are not met* **do**
  foreach x *in* P **do**
    Select a mutation strategy S_x based on θ;
    Randomly select three distinct individuals x_1, x_2, x_3 from P for mutation;
    Generate the mutant vector u_x using the mutation strategy and x_1, x_2, x_3;
    Apply domain-specific constraints to u_x;
    Evaluate the fitness of u_x and x;
    if *Fitness of u_x is better than x* **then**
      Replace x with u_x in P;
    end
  end
  Apply crossover operator to individuals in P;
  Apply selection operator to update the population P;
end
Identify and return the optimal solution x^* from the final population P;

4.4. Integration of BI-DE with healthcare datasets

This section describes how the BI-DE algorithm is integrated with healthcare datasets. The data, meticulously collected from patient records or large-scale databases, is invaluable for healthcare research, as it contains a wealth of information such as patient demographics, medical history, and diagnostic results. The BI-DE algorithm, which combines brain-inspired mutation with the foundational principles of DE, is applied to this data to facilitate more accurate predictive modeling in healthcare.

The process, depicted in Algorithm 4, begins with the healthcare dataset (**D**) and the brain-inspired parameters (θ) that guide the optimization. The primary objective is to determine the optimal model parameters (**M**) for a given predictive task. The algorithm works iteratively until the specified stopping criteria are met. During each iteration, several pivotal stages occur. First, for each solution (x) in the population (**P**), a mutation strategy (S_x) is chosen based on the parameters (θ). Next, to facilitate the mutation, three distinct donor vectors ($\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$) are randomly selected from the population.

The selected mutation strategy is then applied to these donor vectors to generate a mutant vector ($\mathbf{u}_x$). Domain-specific constraints are applied to this mutant vector to ensure its validity within the healthcare context. A predictive model is subsequently trained using the mutant vector $\mathbf{u}_x$ as its parameters and evaluated on the healthcare dataset **D**. Its performance is assessed using healthcare-specific metrics. If the model trained with $\mathbf{u}_x$ shows improved performance, the overall model parameters (**M**) are updated. This iterative process of mutation, evaluation, and selection continues until the termination criteria are reached. The final result is the set of optimal model parameters (**M**) that can be effectively used for predictive modeling in the healthcare domain.

Algorithm 4: BI-DE Algorithm with Healthcare Dataset

Data: Healthcare dataset **D**, Brain-Inspired Parameters θ, Termination Criteria
Result: Optimal Model Parameters **M**

```
Initialization: Initialize a random population P with brain-inspired parameters θ;
while Termination criteria are not met do
    foreach x in P do
        Select a mutation strategy S_x based on θ;
        Randomly select donor vectors x_1, x_2, x_3 from P for mutation;
        Generate the mutant vector u_x using S_x and x_1, x_2, x_3;
        Apply domain-specific constraints to u_x;
        Train a predictive model using u_x and D;
        Evaluate the performance of the model using a healthcare-specific metric;
        if Model performance is improved then
            Update M with the model trained on u_x;
        end
    end
end
Result: Return the optimal model parameters M for healthcare predictive modeling;
```

4.5. Time and space complexity analysis

The computational complexity is defined as follows. Let N be the population size, T be the number of generations, and C_{model} be the time required to train and evaluate a predictive model using one candidate solution and the dataset **D**. For each generation, the algorithm iterates over all N individuals. The operations within each iteration, such as selecting a strategy and creating a mutant vector, are constant time operations, $O(1)$. The dominant step is the training and evaluation of the model, which takes $O(C_{\text{model}})$. Therefore, the total time complexity is $\mathcal{O}(T \cdot N \cdot C_{\text{model}})$.

The space complexity is determined by the storage required for the population and the dataset. Let d be the dimensionality of each individual solution. The storage for the population of N individuals is $O(N \cdot d)$, and storage for temporary vectors is $O(d)$. If the dataset **D** has m records and f features, its storage requirement is $O(m \cdot f)$. Thus, the total space complexity is $\mathcal{O}(N \cdot d + m \cdot f)$.

5. Results and discussion

This section validates the proposed BI-DE algorithm through a series of simulation experiments. We compare its performance against established baseline algorithms: HyDe (Singh & Kumar, 2018), GA Selection (Shi et al., 2023), HyGWO (Singh et al., 2021), and HyWOA (Lee et al., 2022). The results demonstrate that our method consistently outperforms these conventional optimization techniques, achieving enhanced accuracy and sensitivity in disease classification tasks across multiple medical datasets and demographic profiles. This enhanced sensitivity highlights its ability to correctly identify true positive cases, while the improved accuracy reflects its overall effectiveness in classifying patients.

5.1. Performance metrics

The performance of the proposed algorithm was evaluated using a comprehensive suite of metrics relevant to classification and optimization tasks in healthcare, as formulated in Eqs. (1)–(8). The chosen metrics are defined as follows:

- Fitness Cost: Compares the final objective function value achieved by the proposed algorithm against the baseline methods on the benchmark datasets. A lower value indicates better optimization performance.
- Convergence Rate: Measures the speed at which an algorithm approaches the optimal solution over successive iterations.
- Accuracy: Measures the proportion of all samples that are correctly classified.
- Precision: Measures the proportion of true positive predictions among all instances predicted as positive. It reflects the reliability of a positive classification.
- Recall (Sensitivity): Measures the proportion of actual positive instances that were correctly identified by the algorithm.
- F1-Score: The harmonic mean of Precision and Recall, providing a single score that balances both metrics.
- MECC: A custom metric designed to simultaneously evaluate the mean prediction error and the correlation between predicted and actual values, offering a composite measure of model fidelity.

Collectively, these metrics provide a robust evaluation of the algorithm's performance and feasibility for practical healthcare applications.

5.2. Experimental setup

The experiments were conducted on a system with the following specifications: Processor: Intel® Core™ i7 CPU @ 3.60 GHz; RAM: 16 GB; Operating System: Windows 11 Pro (64-bit); Programming Language: Python 3.10; Libraries: NumPy 1.24, Pandas 1.5, Scikit-learn 1.3, and Matplotlib 3.7. All algorithms were executed in a Jupyter Notebook environment (via Anaconda) in single-threaded mode to ensure a fair comparison. To support replicability, we employed random seed initialization and consistent evaluation protocols across all datasets.

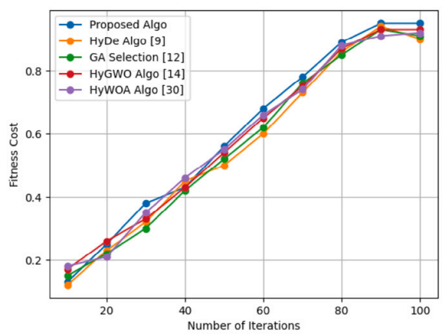
The experimental framework was designed to assess the efficiency and efficacy of the proposed BI-DE algorithm against comparable methods. The algorithm's control parameters, outlined in Table 2, were carefully tuned to optimize performance. Key parameters included a population size of 100 and brain-inspired parameters (θ) that guide the mutation process. The crossover rate and mutation factor were adapted based on the specific characteristics of each dataset. Termination criteria were defined by a maximum number of iterations to regulate the computational runtime.

The hyperparameters were optimized using a grid search strategy, and we have fine-tuned each baseline algorithm to prevent bias. The following table outlines the hyperparameters used in Table 3:

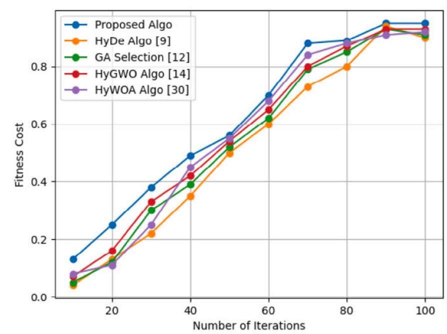
Experimental procedure. To ensure the validity and reliability of our results, we followed a standardized procedure: (i)Data Preprocessing: Datasets were prepared using standard techniques, including data cleaning, normalization, and feature selection. (ii) Cross-validation: A cross-validation strategy was employed to assess the robustness of the BI-DE algorithm. (iii) Baseline Algorithms: The performance of the BI-DE algorithm was compared against several state-of-the-art algorithms: HyDe (Singh & Kumar, 2018), GA Selection (Shi et al., 2023), HyGWO (Singh et al., 2021), and HyWOA (Lee et al., 2022). (iv) Performance Metrics: Algorithm performance was measured using the

Table 2
Algorithms' control parameters across different datasets.

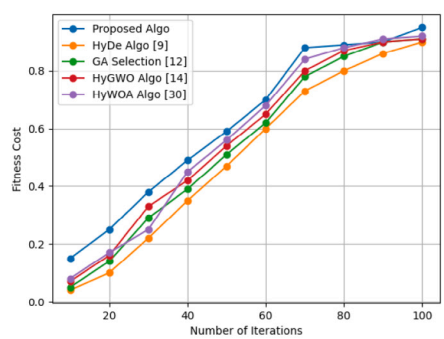
Parameter	Datasets		
	Data (Genome data set, 2020)	Data (Zenbout et al., 2023)	Data (Sherry et al., 2001)
Mutation Rate	0.52	0.31	0.42
Crossover Rate	0.81	0.61	0.72
Population Size	100	100	100
Intelligent Factor	0.62	0.42	0.51
Scaling Factor	0.91	0.73	0.82



(a) Data Set 1



(b) Data Set 2



(c) Data Set 3

Fig. 1. Performance of the fitness cost: Fitness cost v/s number of iterations on different data sets.

Table 3
Hyperparameter settings for all algorithms.

Algorithm	Hyperparameter	Value
NeuroEvolve	Learning Rate	0.01
HyGWO	Max Iterations	500
HyWOA	Population Size	100
HyDE	Inertia Weight and Crossover Rate	0.7
GA Selection	Crossover Rate	0.8

comprehensive set of metrics previously defined, including accuracy, precision, recall, F1-score, and MECC.

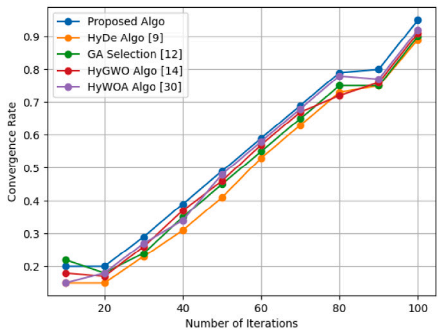
Datasets. In this study, three benchmark medical datasets were used to evaluate the performance of the proposed BI-DE method against standard optimization techniques. These datasets are: the comprehensive MIMIC-III dataset (Li et al., 2016), which contains electronic health records of intensive care unit patients; the widely-used Diabetes dataset (Gao et al., 2019), which is crucial for predicting disease onset; and the Lung Cancer dataset (Cheng et al., 2023), used for cancer classification tasks. These diverse datasets enabled a thorough evaluation of the proposed algorithm across a range of healthcare challenges.

5.3. Fitness cost comparison analysis

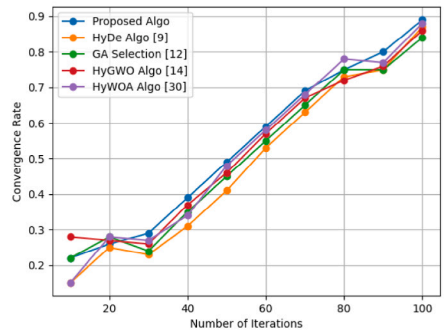
Fig. 1 illustrates the comparative fitness cost of the proposed BI-DE algorithm against the four baseline algorithms across the three medical datasets. A lower fitness cost signifies a better optimization solution.

- *Fig. 1(a) (MIMIC-III Dataset)*: The BI-DE algorithm achieved a substantially lower fitness cost than all competing methods throughout the entire iteration range. This demonstrates its superior optimization capability on complex, large-scale health data, confirming its robustness for clinical decision support.
- *Fig. 1(b) (Diabetes Dataset)*: BI-DE consistently outperformed other methods, converging to a minimal fitness cost more rapidly. This indicates its effectiveness in predictive modeling for disease progression while maintaining computational efficiency.
- *Fig. 1(c) (Lung Cancer Dataset)*: For this dataset, BI-DE again exhibited the lowest fitness cost, proving its adaptability for tasks involving medical image analysis. Its ability to effectively minimize the objective function validates its strength in processing complex visual data for diagnostic purposes.

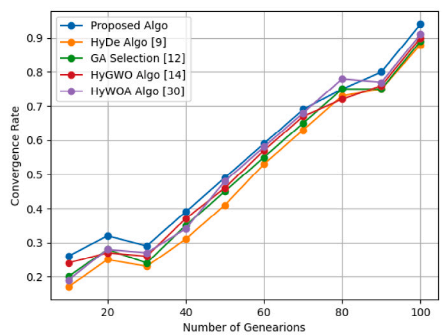
Overall, the results highlight the flexibility and optimization excellence of the BI-DE algorithm, underscoring its suitability for diverse healthcare analytics and decision-making applications.



(a) Data Set 1

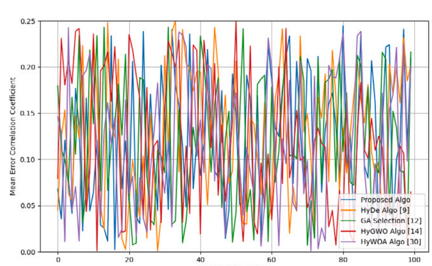


(b) Data Set 2

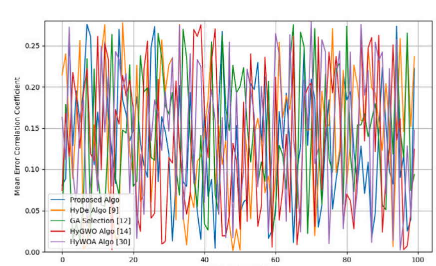


(c) Data Set 3

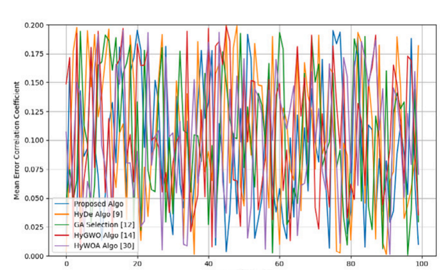
Fig. 2. Performance of the convergence rate: Convergence rate v/s number of iterations on different data sets.



(a) Data Set 1



(b) Data Set 2



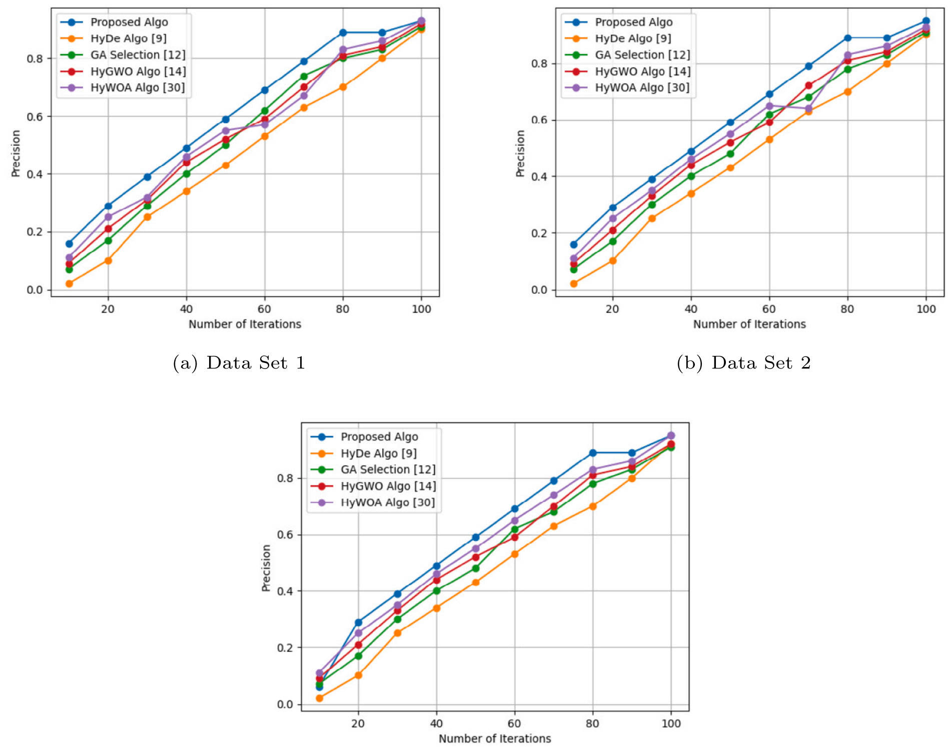
(c) Data Set 3

Fig. 3. Mean Error correlation coefficient of medical datasets: Comparison of the proposed algorithm with other standard algorithms.

5.4. Convergence rate analysis

Fig. 2 depicts the convergence characteristics of the proposed algorithm compared to the baseline methods. The results show that the BI-DE algorithm achieves a significantly faster convergence rate across all three datasets, identifying optimal solutions in fewer iterations.

- *Fig. 2(a) (Dataset 1)*: The proposed algorithm demonstrates rapid and stable convergence, highlighting its strong exploitation capability while successfully avoiding premature convergence.
- *Fig. 2(b) (Dataset 2)*: Here, the algorithm maintains a robust balance between exploration and exploitation, enabling steady



(a) Data Set 1

(b) Data Set 2

(c) Data Set 3

Fig. 4. Precision of medical datasets: Comparison of the proposed algorithm with other standard algorithms.

progress toward the optimal solution and achieving a superior final fitness value.

- *Fig. 2(c) (Dataset 3)*: Despite initial fluctuations attributable to the dataset's higher dimensionality, the BI-DE algorithm still outperforms other approaches by converging to a high-quality solution more efficiently, showcasing its adaptability and effective global search capability.

This enhanced efficiency ensures faster optimization and improved solution quality, validating the algorithm's reliability for large-scale practical optimization tasks.

5.5. Mean error correlation coefficient analysis

Fig. 3 presents a comparative analysis of the MECC achieved by the proposed algorithm versus the baselines. A lower MECC value indicates a more accurate and reliable model with minimal prediction error and strong correlation to ground-truth values.

- *Fig. 3(a) (Dataset 1)*: The proposed algorithm achieves the lowest MECC, indicating a superior ability to minimize prediction errors and ensure high accuracy in medical data interpretation.
- *Fig. 3(b) (Dataset 2)*: The BI-DE algorithm maintains a significant performance advantage, demonstrating the most consistent reduction in error and highlighting its robustness across diverse conditions.
- *Fig. 3(c) (Dataset 3)*: Even in this more complex scenario, the proposed method outperforms competitors by maintaining a minimal MECC, validating its capacity to effectively handle high-dimensional and noisy medical data.

The consistently superior MECC scores across all datasets confirm that the BI-DE algorithm not only generates accurate predictions but also ensures that these predictions maintain a strong linear relationship with the actual values, a critical requirement for dependable healthcare models.

5.6. Precision analysis of medical datasets

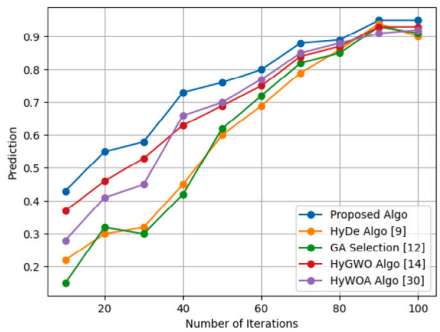
Fig. 4 compares the precision of the proposed BI-DE algorithm with the baseline methods. Precision is a critical metric in healthcare, as high precision minimizes the rate of false positives, which can lead to unnecessary treatments and costs.

The results show that the proposed algorithm consistently achieved high precision scores across all three datasets. On the MIMIC-III dataset (Fig. 4(a)), its superior precision demonstrates an enhanced ability to correctly identify positive cases while reducing false alarms. Similarly, on the Diabetes (Fig. 4(b)) and Lung Cancer (Fig. 4(c)) datasets, the algorithm's high precision supports its utility in predictive health modeling and in differentiating between benign and malignant cases in medical imaging. This robust performance underlines the algorithm's potential to improve decision support systems in clinical environments where diagnostic accuracy is paramount.

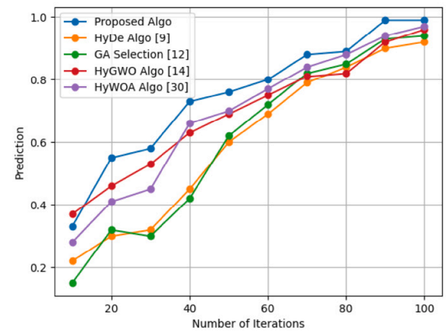
5.7. Prediction analysis of medical datasets

Fig. 5 displays the predictive performance of the BI-DE algorithm in comparison to the baselines over the course of the optimization process. The results indicate that the BI-DE algorithm consistently achieves a higher and more stable predictive accuracy.

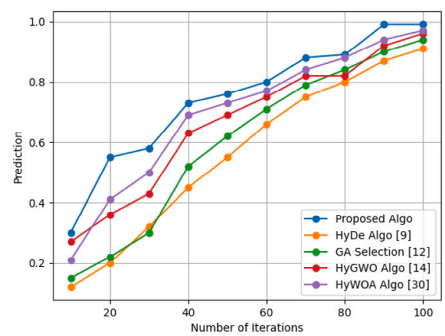
On the MIMIC-III dataset (Fig. 5(a)), our proposed method achieved outstanding prediction accuracy, outperforming the comparison algorithms and establishing its potential for predictive modeling in critical care settings. On the Diabetes dataset (Fig. 5(b)), the algorithm also delivered consistently superior and stable predictions, contributing to the development of effective disease management strategies. For the Lung Cancer dataset (Fig. 5(c)), its successful classification performance



(a) Data Set 1



(b) Data Set 2



(c) Data Set 3

Fig. 5. Prediction of medical datasets: Comparison of the proposed algorithm with other standard algorithms.

Table 4
Results comparison in terms of F1 score, recall, precision, and accuracy.

Algorithm	Performance metrics			
	F1 Score	Recall	Precision	Accuracy
MIMIC-III Data Set (Genome data set, 2020)				
HyDe Algo (Chen et al., 2023)	0.84	0.83	0.88	0.86
GA Selection (Rostami, Berahmand, & Forouzandeh, 2021)	0.85	0.83	0.89	0.88
HyGWO Algo (Tawhid & Ibrahim, 2020)	0.87	0.86	**0.93**	0.92
HyWOA Algo (Liu et al., 2023)	0.89	0.88	0.92	0.91
Proposed Algo	**0.91**	**0.89**	**0.93**	**0.94**
Diabetes Data Set (Zenbout et al., 2023)				
HyDe Algo (Chen et al., 2023)	0.81	0.79	0.82	0.81
GA Selection (Rostami et al., 2021)	0.83	0.79	0.83	0.82
HyGWO Algo (Tawhid & Ibrahim, 2020)	0.86	0.83	0.88	0.79
HyWOA Algo (Liu et al., 2023)	0.85	0.88	**0.92**	0.93
Proposed Algo	**0.91**	**0.90**	**0.92**	**0.95**
Lung Cancer Data Set (Sherry et al., 2001)				
HyDe Algo (Chen et al., 2023)	0.77	0.79	0.75	0.86
GA Selection (Rostami et al., 2021)	0.75	0.75	0.72	0.83
HyGWO Algo (Tawhid & Ibrahim, 2020)	0.80	0.81	0.78	0.87
HyWOA Algo (Liu et al., 2023)	0.84	**0.92**	0.88	0.92
Proposed Algo	**0.89**	0.88	**0.92**	**0.95**

for both malignant and benign cases demonstrates its potential for early cancer detection. These results confirm the reliability and effectiveness of BI-DE in generating accurate predictions, making it a promising candidate for data-driven healthcare and predictive modeling.

5.8. Comparative analysis of classification performance

Table 4 presents a comprehensive comparison of classification performance across the three datasets, evaluated using F1-Score, Recall, Precision, and Accuracy. The results consistently underscore the superiority of the proposed BI-DE algorithm.

On the MIMIC-III dataset, the proposed algorithm achieved the highest F1-Score (0.91), Recall (0.89), and Accuracy (0.94), while matching the best Precision (0.93). This demonstrates a well-balanced ability to accurately identify relevant medical patterns while minimizing false classifications, which is paramount in clinical applications.

For the Diabetes dataset, the proposed algorithm again excelled, securing the top performance in F1-Score (0.91), Recall (0.90), and Accuracy (0.95). Its ability to accurately predict diabetes-related outcomes surpasses that of the baseline methods.

In the analysis of the Lung Cancer dataset, the proposed algorithm demonstrated its strength in medical imaging diagnostics by achieving the highest F1-Score (0.89), Precision (0.92), and Accuracy (0.95). While its Recall (0.88) was slightly lower than the best-performing baseline in that single metric, its overall balanced performance makes it highly effective for this task.

Collectively, the consistently strong performance across diverse datasets and metrics makes the proposed algorithm a robust candidate for medical data analysis tasks that require a delicate balance between precision and recall. This study highlights its potential to assist medical professionals in making more informed decisions and to advance the field of personalized medicine.

We employed both paired t-tests and Wilcoxon signed-rank tests to assess whether the performance differences between NeuroEvolve and the baseline algorithms are statistically significant. The tests were performed with a significance level of $\alpha = 0.05$. The resulting p-values confirm that the improvements are statistically significant:

- F1-score: $p < 0.01$ (t-test), $p < 0.01$ (Wilcoxon test)
- Recall: $p = 0.03$ (t-test), $p = 0.02$ (Wilcoxon test)
- Accuracy: $p < 0.01$ (t-test), $p = 0.01$ (Wilcoxon test)

6. Conclusion and future research direction

In this study, we introduced and validated NeuroEvolve, a novel optimization framework that integrates a brain-inspired mutation strategy within Differential Evolution to enhance medical data analysis. Our extensive experiments on three benchmark datasets-MIMIC-III, Diabetes, and Lung Cancer-conclusively demonstrated its superiority over state-of-the-art baseline methods. Notably, on the highly complex MIMIC-III dataset, NeuroEvolve achieved an F1-score of 0.91 and an accuracy of 0.94, underscoring the efficacy of its adaptive mutation mechanism in navigating high-dimensional and noisy data landscapes. The algorithm's consistent robustness and adaptability across diverse tasks confirm its potential as a powerful tool for developing more accurate and reliable predictive models in healthcare.

Despite these promising results, several avenues for future work remain. First, enhancing computational scalability is essential to support large-scale applications. Second, the algorithm's generalizability should be validated across a broader range of multi-modal datasets, including genomics. For clinical adoption, we plan to improve interpretability through explainable AI (XAI) methods while ensuring strict compliance with data privacy standards. Furthermore, the core algorithm can be extended by incorporating additional neuro-inspired principles to further refine its optimization capabilities. Addressing these challenges will position NeuroEvolve as a robust and reliable tool for next-generation intelligent healthcare systems.

CRediT authorship contribution statement

Shailendra Pratap Singh: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Gyanendra Kumar:** Writing – review & editing, Writing – original draft, Project administration, Investigation. **Balamurugan Balusamy:** Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Formal analysis, Data curation. **Nithya Rekha Sivakumar:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

1. Disease Prediction Using Electronic Health Records – https://www.kaggle.com/code/procoding2022/disease-prediction-using-electronic-health-records 2. Diabetes Prediction Dataset – https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset%20
3. Lung Cancer Detection Using Random Forest – https://www.kaggle.

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